

Project Description

RUSL FOT Academic Companion - RUSLearn

Mobile Application Development - ICT 3219
Bachelor of Information and Communication Technology (BICT) Degree Program

Department of Information and Communication Technology
Faculty of Technology

Rajarata University of Sri Lanka

Mihintale.

Group Details

Student Name	Index Number	Registration Number
W.W.V.D.S.Perera	2065	ITT/2022/085
G.D.M.Gurusingha	2017	ITT/2022/036
S.A.Kalupahana	2030	ITT/2022/050
G.R.M.H.Gunawardhana	2016	ITT/2022/035
M.K.P.Sachintha	2075	ITT/2022/095

Table of Contents

1 Project Overview 1
2 Problem Statement 1
3 Proposed Solution 2
4 Functional Requirements 2
5 Non-Functional Requirements 3
6 Technology Stack 4
7 Expected Outcome 4

1 Project Overview

The FOT Academic Companion is an Android-based mobile application designed to help students of the Faculty of Technology (FOT) access academic resources and manage their academic progress from one centralized platform.

The application will provide easy access to past examination papers organized by department, subject, semester, academic year, and examination year. Students will be able to search, filter, view, download, and save papers for later use.

In addition to past papers, the application will provide academic management tools such as a GPA and CGPA calculator, GPA prediction, subject tracking, semester information, credit tracking, academic progress monitoring, and a personal dashboard. These features will help students understand their academic status and keep track of the subjects they have completed, are currently taking, and may need to complete in future semesters.

The mobile application will communicate with a backend service and database to retrieve subjects, examination papers, and other required information. Administrative content management will be handled through a separate web-based system and is outside the scope of the student mobile application.

2 Problem Statement

Students often use different sources such as WhatsApp groups, Google Drive folders, social media groups, personal collections, and online platforms to find past examination papers and other academic information. Because these resources are spread across different locations, students may spend unnecessary time searching for the correct paper or information.

Students also need to keep track of their academic progress throughout their degree. It can be difficult to clearly understand their current GPA or CGPA, estimate the effect of possible grades on their GPA, remember which subjects they have already completed, and identify subjects they are currently studying or may need to take in future semesters. Academic information can therefore become difficult to organize and follow over time.

Although an existing web-based past-paper portal provides access to examination papers, it does not by itself provide a unified mobile experience for both academic resources and academic progress tracking. Students who mainly use smartphones would benefit from a simple application that brings these commonly needed functions together in one place.

Therefore, there is a need for a user-friendly mobile application that combines past paper access with GPA tools, subject tracking, semester information, and academic progress monitoring in a centralized platform.

3 Proposed Solution

The proposed system is an Android mobile application that acts as an academic companion for FOT students. The application will combine past examination paper access and academic progress tools in a single mobile platform.

The past-paper section will allow students to browse and search papers using department, subject, semester, academic year, and examination year. Students will be able to view papers, download them for later access, and save frequently used papers as favourites.

The academic section will allow students to enter their subjects, credits, and grades and calculate their semester GPA and overall CGPA. The GPA prediction feature will allow students to test possible grades and understand how they may affect their GPA. The application will also allow students to maintain a record of completed, current, and upcoming subjects and view their progress based on completed and remaining credits.

A personal dashboard will provide a simple overview of the student's academic status, such as GPA, CGPA, current semester, subject progress, and saved resources. Paper requests and offline access to downloaded papers may also be provided to improve convenience.

The mobile application will use a backend API to retrieve and update required student-related data and academic resources. Administrative operations such as adding, updating, or removing papers will be performed through a separate web administration system and will not be part of the mobile application.

4 Functional Requirements

1. **FR01 – User Authentication:** The system shall allow students to register and log in using valid credentials.
2. **FR02 – Browse Past Papers:** The system shall allow students to browse past examination papers by department, subject, semester, academic year, and examination year.
3. **FR03 – Search Past Papers:** The system shall allow students to search for papers using subject names, subject codes, and relevant keywords.
4. **FR04 – Filter Papers:** The system shall allow students to filter examination papers using criteria such as department, semester, academic year, and examination year.
5. **FR05 – View and Download Papers:** The system shall allow students to view available examination papers in PDF format and download papers for later access.
6. **FR06 – Save Favourite Papers:** The system shall allow students to save examination papers as favourites and access them from a dedicated section.
7. **FR07 – GPA and CGPA Calculation:** The system shall allow students to enter subjects, credits, and grades and calculate semester GPA and cumulative GPA.
8. **FR08 – GPA Prediction:** The system shall allow students to enter possible future grades and estimate the effect on their GPA or CGPA.
9. **FR09 – Subject Tracking:** The system shall allow students to record and view completed, current, and upcoming subjects.

10. **FR10 – Semester Information:** The system shall allow students to view subjects and academic information organized by semester.
11. **FR11 – Academic Progress Tracking:** The system shall allow students to track completed credits, remaining credits, and overall academic progress.
12. **FR12 – Personal Dashboard:** The system shall provide a dashboard showing key academic information such as GPA, CGPA, current semester, subject status, credit progress, and recent or saved resources.
13. **FR13 – Paper Requests:** The system shall allow students to submit a request when a required examination paper is not available in the repository.
14. **FR14 – Offline Access:** The system shall allow students to access examination papers that have been downloaded to the device without an active internet connection.

5 Non-Functional Requirements

1. **NFR01 – Usability:** The application shall provide a simple, clear, and mobile-friendly interface so that students can complete common tasks with minimal steps.
2. **NFR02 – Performance:** The application shall return search results, GPA calculations, and other requested information within a reasonable response time under normal conditions.
3. **NFR03 – Reliability:** The application shall handle common errors such as network failures, unavailable documents, and invalid input without crashing.
4. **NFR04 – Security:** User credentials and personal information shall be protected using appropriate authentication and secure communication methods.
5. **NFR05 – Availability:** The application shall be available when the required backend services and network connection are available. Downloaded papers shall remain accessible offline.
6. **NFR06 – Compatibility:** The application shall operate on supported Android smartphones with different screen sizes and resolutions.
7. **NFR07 – Maintainability:** The application shall use a structured and modular architecture so that features can be maintained and extended easily.
8. **NFR08 – Scalability:** The system should support an increasing number of students, subjects, and examination papers without significant performance degradation.
9. **NFR09 – Data Accuracy:** The application shall perform GPA, CGPA, credit, and progress calculations consistently according to the defined academic rules used by the application.

6 Technology Stack

- Programming Language: Java
- IDE: Android Studio
- Platform: Android
- UI: XML
- Backend Communication: REST API
- Data Format: JSON
- Database: PostgreSQL
- Document Format: PDF
- Version Control: Git/GitHub

7 Expected Outcome

The project is expected to produce a reliable and user-friendly Android application that gives FOT students a centralized way to access past examination papers and manage important academic information.

The application will reduce the time and effort required to find examination papers while also helping students understand their GPA and CGPA, explore possible GPA outcomes, keep track of completed, current, and upcoming subjects, and monitor their credit and academic progress.

The final system will provide a practical mobile academic companion that brings commonly used student functions together in one application while keeping administrative content management separate from the student-facing mobile app.